

# Module 04: Cognition — Thinking as Embodied Presence

## Learning Objectives

1. Define **cognition** as an embodied, enactive process inseparable from the living body's ongoing self-regulation and present-moment awareness.
2. Explain how thoughts arise as **predictions** within the generative model, carrying somatic signatures that reveal their embodied origin.
3. Analyze the relationship between **mindful meta-cognition** and precision weighting in active inference, showing how present-moment awareness transforms the cognitive landscape.
4. Apply the concept of **cognitive decentering** to understand how living presence creates a gap between thought and reactive action.

## Introduction

Sit quietly and wait for a thought to arrive. Notice that you do not choose it — it appears, unbidden, like a bubble rising through water. Now notice something subtler: the thought has a feeling. "I need to call my mother" arrives with a slight tug in the chest. "What if I fail?" comes with a tightening in the throat. "This is peaceful" brings a softening behind the eyes. Cognition, experienced from within the living present, is never disembodied. Every thought has a somatic texture, a felt quality that reveals its roots in the body's predictive machinery.

The Western philosophical tradition long treated cognition as the mind's private theater — Descartes' *res cogitans*, wholly separate from the body. The enactive tradition overturns this separation. Francisco Varela, Evan Thompson, and Eleanor Rosch (1991) argued that cognition is not the manipulation of abstract symbols but the embodied activity of a living system coupled to its world. Humberto Maturana insisted that "everything said is said by someone" — every cognitive act is the act of a living body, situated in time and place. Active inference formalizes this insight: thoughts are predictions generated by a hierarchical

generative model whose deepest layers are visceral, autonomic, and interoceptive. To think is to predict, and to predict is to be a body anticipating its own future states.

## **Key Concepts**

### **1. Cognition as Embodied Prediction**

In the active inference framework, cognition is not a separate faculty layered on top of perception and action but an integral part of the same predictive hierarchy. Thoughts are high-level predictions — hypotheses about the causes of sensory states and the likely consequences of actions. What makes cognition embodied is that these predictions are generated by a model whose parameters have been shaped by a lifetime of bodily interaction with the world. When you think "this room is cold," the thought is not an abstract proposition but a prediction arising from interoceptive signals (goosebumps, vasoconstriction, shivering readiness) that the generative model interprets as temperature deviation from homeostatic set-points.

Evan Thompson (2007) emphasizes that this embodied view of cognition dissolves the boundary between "higher" and "lower" cognitive processes. Abstract reasoning, memory retrieval, and imaginative simulation all draw on the same predictive architecture that governs breathing, balance, and thermoregulation. The difference is one of hierarchical depth, not of kind.

### **2. The Somatic Signature of Thought**

Antonio Damasio's somatic marker hypothesis proposes that every cognitive evaluation is accompanied by a bodily state — a somatic marker — that colors the thought with emotional valence. From an active inference perspective, somatic markers are interoceptive prediction errors associated with particular cognitive hypotheses. When the generative model entertains the prediction "I will lose my job," it simultaneously generates predictions about the bodily states associated with that scenario: cortisol release, increased heart rate, muscular tension. These predicted states are felt as anxiety, and they function as precision signals that modulate the salience of the thought.

Mindful awareness of these somatic signatures transforms cognition. Rather than being carried along by the thought, the practitioner can attend to its bodily accompaniment — noticing the

tightness, the heaviness, the acceleration of the heartbeat — and thereby gain insight into the embodied origins of what seemed like purely mental events. This is what contemplative traditions call "thinking about thinking" and what active inference formalizes as hierarchical meta-cognition: the generative model observing its own predictions.

### 3. Cognitive Decentering and Present-Moment Awareness

One of the most significant findings in contemplative science is the phenomenon of **cognitive decentering** — the capacity to observe thoughts as mental events rather than truths about the world. In mindfulness-based cognitive therapy (MBCT), this capacity is the primary mechanism of therapeutic change. Patients learn to notice "I am having the thought that I am worthless" rather than simply believing "I am worthless."

Active inference provides a formal account of decentering. Normally, the generative model treats its own high-level predictions as veridical — the thought is experienced as reality. Decentering involves adding a meta-cognitive layer that assigns reduced precision to thought-content while increasing precision on the process of thinking itself. The prediction error shifts from "the world is threatening" to "my model is generating a threat-prediction." This re-weighting is precisely what present-moment awareness accomplishes: by anchoring attention in current bodily experience (breath, posture, sensation), the practitioner creates a stable reference frame from which thoughts can be observed without being mistaken for reality.

### 4. Counterfactual Cognition and Temporal Presence

Much of human cognition involves counterfactual simulation — imagining scenarios that are not currently the case. Rumination (replaying the past) and worry (rehearsing the future) are forms of counterfactual inference that detach the generative model from present sensory evidence. In active inference terms, these states involve excessive reliance on deep temporal priors at the expense of current sensory likelihood — the model is predicting states that do not correspond to any ongoing sensory flow.

Living presence counteracts this tendency by increasing the precision on present-moment sensory signals. When the meditator notices she has been lost in worry, she returns attention to the breath — re-anchoring the generative model in current interoceptive evidence. This is not the suppression of thought but its contextualization: the counterfactual prediction is

recognized as a prediction rather than experienced as present reality. Varela's concept of the "specious present" — the 2-3 second window of lived temporal experience — suggests that healthy cognition maintains a balance between counterfactual depth and present-moment grounding.

## Active Inference Connection

Cognition in the context of living presence reveals something fundamental about the active inference framework: the generative model is not merely a passive predictor but a self-observing system capable of modulating its own inferential dynamics. When mindful awareness is directed toward the process of thinking itself, the model gains the capacity to adjust precision at the meta-cognitive level — reducing the grip of habitual thought patterns, increasing sensitivity to novel information, and maintaining flexible engagement with the present moment. This is free energy minimization at its most refined: not eliminating prediction error but developing the precision architecture to respond to it with wisdom rather than reactivity.

## Experiential Applications

- **Practice — Thought Labeling Meditation:** Sit in a comfortable position and anchor attention on the breath. When a thought arises, silently label it — "planning," "remembering," "judging," "fantasizing" — and notice the somatic signature that accompanies it. Where in the body does the thought register? What is its felt texture — heavy, buzzing, contracting, expanding? After labeling and sensing, gently return to the breath. This practice develops meta-cognitive precision: the capacity to observe the generative model's predictions without being captured by their content.
- **Case Study — Mindfulness-Based Cognitive Therapy for Depression:** In MBCT, patients who have experienced recurrent depression learn to detect early warning signs — not through intellectual self-monitoring but through embodied awareness. A sinking sensation in the chest, a heaviness in the limbs, a narrowing of visual attention — these somatic markers signal the activation of depressive cognitive patterns before the full syndrome develops. By recognizing these bodily predictions early, patients can deploy strategies (breath awareness, gentle movement, compassionate self-talk) that recalibrate

the generative model before maladaptive cognition takes hold. Clinical trials show that MBCT reduces relapse rates by approximately 50% in patients with three or more previous episodes.

## Cross-References

- **Module 03 (Perception):** Provides the perceptual foundation on which cognition builds — thoughts are higher-order interpretations of perceptual predictions.
- **Module 05 (Action):** Connects cognitive predictions to their motor consequences — how thinking prepares and shapes embodied action.
- **Unit 01 (Felt Sense):** Deepens the interoceptive dimension of embodied cognition through attention to visceral and somatic feeling.
- **Unit 03 (Intuitive Knowing):** Explores how cognition can operate beneath the threshold of reflective awareness as embodied expertise.

## Summary

Cognition, experienced through the lens of living presence, is revealed as an embodied, somatic, and inherently predictive process. Thoughts are not disembodied abstractions but predictions generated by a generative model rooted in the body's ongoing self-regulation. Every thought carries a somatic signature — a felt quality that betrays its visceral origins. Present-moment awareness transforms the cognitive landscape by enabling meta-cognitive observation: the capacity to witness thoughts as predictions rather than truths, to sense their bodily texture, and to maintain flexible engagement with the ongoing flow of experience. Active inference formalizes what contemplative traditions have long taught — that the deepest cognitive freedom comes not from thinking different thoughts but from changing one's relationship to thinking itself.